



Association of Zoos & Aquariums

2023 AZA CONSERVATION GRANTS FUND

Application Instructions

DEADLINE

This 2023 Conservation Grants Fund (CGF) electronic application must be used for all proposals. Completed applications must be submitted via email no later than **5:00 p.m. EDT** on **15 March 2023**. An auto-response email with a time-stamp will confirm receipt of the application.

*Incomplete applications, or applications received after 5:00 p.m. EDT on 15 March 2023, **regardless of cause**, will NOT be eligible for consideration. NO DEADLINE EXCEPTIONS WILL BE MADE.*

MEMBERSHIP

The Principal Investigator (PI) and all Co-Investigators (CIs) *must be individual AZA members* at the Professional Associate, Professional Affiliate, Professional Fellow, or Student level. Institutional membership does not count as individual membership and individual members at the Friend level are not eligible. Other participants actively involved in the completion of the project should be listed as collaborators and are not required to be AZA members. To apply for membership, visit <https://www.aza.org/individual-membership>.

FUNDING TIMELINE

All CGF projects must be completed within 12 months. CGF grants are intended to serve as seed money to attract additional funding and proposals should request funds only for this one-year period. Identical components of a project will not be funded for more than two grant cycles; however, funding requests for different components of the same project may be applied for in subsequent grant cycles.

CGF proposals undergo a two-tiered review process, outlined at <https://www.aza.org/cgf-selection-process>. Applicants will be notified of funding decisions in early September 2023 and funds will be available 2 October 2023. CGF funds can only be used for proposal expenses incurred on or after the award date. Expenses incurred prior to the award date will not be covered by CGF funds.

APPLICATION GUIDELINES

The application and all supplemental forms are posted online at: <http://www.aza.org/cgf-information-for-applicants/>. These documents contain form fields that must be completed and submitted via email to: CGFapp@aza.org. *All application materials must be in electronic format. No hard copies of any application materials will be accepted.*

It is strongly recommended that you save a copy of your completed application and supplemental forms as a backup. Only completed applications, submitted by the deadline defined above, will be reviewed. Applicants will be informed electronically if their application is incomplete.

APPLICATION INSTRUCTIONS AND CHECKLIST: All completed application materials must be submitted **in one email** by the deadline identified above and include the following:

Application: The application is one document consisting of the three sections outlined below. All sections must be complete in order for the application to be considered complete. The application must be submitted as a Microsoft Word document titled **"Application"** and attached to your Application email.

Applications submitted in Portable Document Format (PDF) will NOT be eligible for consideration. NO EXCEPTIONS WILL BE MADE.

Note that “spell checker” and “track changes” functions cannot be used in this form. AZA strongly recommends that applicants compose the content for their application in a separate document to make use of these features, and then copy and paste the final text into the form fields. When composing in a separate document, be aware of character limitations. If the copied text exceeds the character limit when pasted into a form field, *all text exceeding the character limitation will be cut off and will not appear in the final proposal.* Character limits include spaces.

Also note that certain formatting such as italics, bold, and underlining, is unavailable due to the application format. AZA recommends the use of numbering and spacing to emphasize important points in the text.

- ☒ **Cover Sheet.** Fill in the text fields for the project title, amount requested, and PI and CI information. Choose *two* project categories from the given list and identify the SAFE species that is (are) the focus of the project. Include a project abstract of 1500 or fewer characters.
- ☒ **Project Proposal.** Text fields are provided for responses to questions 1-12. Fields have a limit on the number of characters that may be used to keep the application concise. Character limits, including spaces, are stated following each question. If a question is not applicable to your proposal, write “N/A” in the field but *do not leave any form fields blank*, as this may delay processing of your application.
- ☒ **Budget.** Required budgetary information includes:
 - ☒ Detailed Project Budget. Detail how the budget will be allocated to specific items and indicate which funding source will pay for each item. State the total project budget, the amount requested from CGF, and other financial support applied for or received. Each of these amounts should reflect the one-year duration of the project. Funding decisions are announced in early September and CGF funds are to be used for work from the announcement date forward. Expenses incurred prior to the announcement will not be covered by CGF funds.
 - ☒ Budget Justification. Provide an explanation of why specific major items are essential to the project and reasonable in cost.

Required Attachments: The following must be attached to your Application email:

- ☒ **Curricula Vitae (CV).** A CV is required for each PI and CI listed on the Cover Sheet and must follow the format stated in these guidelines. CVs must not exceed two letter-sized (8.5 by 11 inch) pages each and must be typed in a legible font size (12 point recommended); each side of the paper counts as one page. CVs may be attached for collaborators but are not required. Each CV must be submitted as a separate document entitled, “**CV Investigator’s Last Name.**” CVs *exceeding the maximum length will be cut off at the second page.*
- ☒ **Statement of Institutional Support Forms.** Electronically signed *Statements of Institutional Support* from the CEO(s)/Institutional Director(s) (not the COO, General Curator, Vice President, Dean, or Department Director) of the PI’s institution and **each** CI institution *and* collaborating institution must be attached to the Application email as separate documents titled “**SIS Institution’s Name.**” The *Statement of Institutional Support* form can be found with the rest of the Application materials on the AZA website identified above.

Applicants wishing to include support letters from entities *other than* PI and CI institutions, collaborating institutions may submit such items as appendices.

Letters of support from PI and CI institutions and collaborating institutions will not be accepted in lieu of the *Statement of Institutional Support* form.

- ☒ **Statement of SAFE Species Program Alignment Forms.** Electronically signed *Statements of SAFE Species Program Alignment* from the program leader of relevant SAFE species program(s) must be attached to the Application email as separate documents titled “**SOA SAFE Species Program Name.**” The *Statement of SAFE Species Program Alignment* form can be found with the rest of the Application materials on the AZA website identified above.

Letters of support/endorsement from AZA Programs (Taxon Advisory Groups, Species Survival Plans®, SAFE) are **not** accepted, although coordination with AZA SAFE Programs is expected and should be described in Question 1. To assure program review of relevant submissions, AZA SAFE or Animal Program Leaders may serve as expert reviewers in the application review process and may express their support for a proposal at that time.

- ☐ **Permits.** If permits are required to complete your proposed project, you must attach a copy of each to your Application email. Each permit attached to the Application email must be titled “**Permit #.**” *If you are not able to obtain a required permit before submitting this proposal, you must submit it by 15 June 2023 to ensure award eligibility.*
- ☐ **Proposal Resubmission Form.** If a proposal for this project was previously submitted for AZA grant funding (i.e., CGF or SAFE granting program) and rejected, the revised proposal *must* include a completed *Proposal Resubmission Form* as an attachment on the Application email. This attachment must be submitted as a Microsoft Word document titled “**Resubmission Form**” and attached to your Application email. *Resubmission forms submitted in Portable Document Format (PDF) will not be accepted.* The form can be found with the rest of the Application materials on the AZA website identified above.

Note: Proposals that fail to provide all items identified in the above checklist in the correct format will not be reviewed.

Non-Required Attachments: The following attachments to your Application email are optional:

- ☒ **Citations (recommended).** Citations/references for statements made in the Application should be included in a separate document titled “**Citations**” and attached to the Application email.
- ☒ **Appendices.** Appendices are allowed, but may only *clarify*, and not *supplement*, information in the narrative. For example, photos or diagrams would be acceptable, but not videos, publications, or SAFE species program plans. Appendices must be attached to the Application email titled “**Appendix #.**”

Note: Appendices that do not adhere to these guidelines will be removed from the proposal prior to review.

SUBMISSION INSTRUCTIONS

By submitting your application, you agree, if funded, to abide by the Grant Terms and Conditions posted on AZA’s website at <http://www.aza.org/cgf-information-for-applicants/>.

Proposals must be submitted electronically using the 2023 Application and related forms via a **single email, with the proposal title as the subject line**, to CGFapp@aza.org no later than 5:00 p.m. EDT on 15 March 2023. Hard copies of any application materials or applications received after this deadline will not be accepted regardless of cause.

AZA's email system accepts messages up to 8 MB in size. In cases when applications are too large to be accepted in one email, AZA will accept the application submission in two separate emails, provided that the same proposal title is identified in the subject line of each.

The 2023 Application email **must** include the following attachments

- Application (this completed form);
- CVs for each Principal and Co- Investigator(s);
- Statements of Institutional Support for each PI and CI institution and collaborating institution;
- Statement(s) of SAFE Species Program Support; and
- A Proposal Resubmission form, if applicable.

The Application email may also include citation and appendices attachments.

Auto-response emails will be sent to applicants confirming receipt of the application package. If an auto-response is not received, please resubmit the application. If an applicant submits more than one proposal, the person should ensure that an auto-response was received for *each* submission. If submission problems arise, contact the AZA office at 301-562-0777. If you submit your application after the 5:00 p.m. deadline, you may still receive an auto-response from the Application inbox; however, your proposal will not be reviewed.

Applications which fail to follow submission instructions or are time-stamped after the deadline will not be considered and applicants will be notified that their application is inadmissible.

APPLICATION ASSISTANCE

The CGF website (<http://www.aza.org/cgf-information-for-applicants/>) contains a wealth of information to help applicants compose a competitive proposal.

Proposals received before 8 March will be reviewed by AZA staff to ensure that the application is completed correctly. Early submission allows time for AZA staff to inform applicants if problems are noted in time for resubmission. Please note that AZA staff does not review early proposals for content.

2023 AZA CONSERVATION GRANTS FUND
Cover Sheet

Project Number:
For AZA use only

Title of Project: A multi-omics approach to understanding teratospermia in cheetahs (*Acinonyx jubatus*)

Amount Requested: \$23,300 (round to the nearest whole dollar)

Project Start Date: 10/10/2023

End Date: 10/09/2024

(CGF funding is for one year only, beginning no earlier than the award date)

Investigators: Each Principal Investigator (PI) and Co-Investigators (CI) *must be an individual AZA member at the time of application* at the Professional Associate, Professional Affiliate, Professional Fellow, or Student level. Other participants should be listed as collaborators on this form.

A Curriculum Vitae (CV) is required for each PI and CI, and a Statement of Institutional Support (SIS) form is required for each PI and CI Organization.

1. Principal Investigator (PI) Name and Title: [REDACTED]

CV Attached? ☒ (required – no more than 2 pages)

Checking this box indicates agreement to the CGF Terms and Conditions, including affirmation that the organization is or will be a program partner of the relevant SAFE species program(s) upon acceptance of an award (required; available at: <http://www.aza.org/cgf-information-for-applicants/>). ☒

Individual AZA Member Number: [REDACTED]

Institution/Organization: [REDACTED]

Statement of Institutional Support Form Attached? ☒ (required)

Institutional Address: [REDACTED]
[REDACTED]

2. Co-Investigator (CI) Name and Title: [REDACTED]

CV Attached? ☒ (required – no more than 2 pages)

Checking this box indicates agreement to the CGF Terms and Conditions, including affirmation that the organization is or will be a program partner of the relevant SAFE species program(s) upon acceptance of an award (available at: <http://www.aza.org/cgf-information-for-applicants/>). ☒ (required)

Individual AZA Member Number: [REDACTED] (required)

Institution/Organization: [REDACTED]

Statement of Institutional Support Form Attached? ☒ (required)

Institutional Address: [REDACTED]

Phone: [REDACTED]

E-mail: [REDACTED]

3. Co-Investigator (CI) Name and Title: [REDACTED]

CV Attached? ☒ (required – no more than 2 pages)

Checking this box indicates agreement to the CGF Terms and Conditions, including affirmation that the organization is or will be a program partner of the relevant SAFE species program(s) upon acceptance of an award (available at: <http://www.aza.org/cgf-information-for-applicants/>). ☒ (required)

Individual AZA Member Number: [REDACTED] (required)

Institution/Organization: [REDACTED]

Statement of Institutional Support Form Attached? ☒ (required)

Institutional Address: [REDACTED]
[REDACTED]

E-mail: [REDACTED]

Information about additional co-investigators may be included in a separate document titled “**Additional Co-Investigators**” and attached to your Application email.

Collaborating Institutions or Organizations: A collaborating institution is defined as any institution actively involved in the design or completion of the project. Funding institutions *not* actively involved in project design or completion should be listed in the “Grant/Matching Support” section of the Project Budget and need not be listed as collaborators.

List all collaborating institutions below, including the institutions of the PI and CIs listed on the Cover Sheet, as well as those associated with the personnel listed in Question 6, and attach a Statement of Institutional Support form for each collaborating institution to the Application email.

Collaborating Institution Names:

Smithsonian's National Zoo and Conservation Biology Institute

Statement of Institutional Support Form

Attached? ☒ (required)

George Mason University

Statement of Institutional Support Form Attached? ☒ (required)

Statement of Institutional Support Form Attached? ☐ (required)

Statement of Institutional Support Form Attached? ☐ (required)

Statement of Institutional Support Form Attached? ☐ (required)

Note: If additional Collaborating Institutions or Organizations need to be included, add their information in a separate document titled “**Additional Collaborators**” and include it as an attachment to your Application email.

Project Categories (choose two that best describe the project):

☐ Animal / Population Management (*Ex situ*)

☐ Conservation Education / Engagement

☒ Animal Health

☐ Field Conservation (*In situ*)

☐ Animal Welfare

☒ Research

☐ Communications / Public Relations

Identify the SAFE species program(s) this project will benefit: Cheetah (*Acinonyx jubatus*)

Project Abstract (limit 1,500 characters with spaces):

Teratospermia – ejaculation containing <40% morphologically normal sperm – is widespread in the male cheetah population and inhibits breeding success. However, we do not fully understand its genetic basis. Cheetahs carry fixed deleterious variants in fertility-associated genes. Their motile sperm output is also associated with levels of human interaction and public exposure, suggesting an environmental influence on the teratospermia phenotype, perhaps via altered gene regulation. To gain greater insight into the genetic and epigenetic bases of teratospermia, we propose to conduct whole-genome sequencing and reduced representation bisulfite sequencing on 30 whole blood and semen samples from captive male cheetahs, including both on-exhibit and off-exhibit animals. Our goal is to determine SNPs and differential methylation sites associated with phenotypic aspects of teratospermia. SNPs will be called using standard pipelines, and their frequencies, effects, and site conservation scores will be determined. DNA methylation status at relevant loci will be compared between on-exhibit, off-exhibit, and control groups; a domestic cat will be the control. Our findings will advance understanding of this longstanding conservation concern in endangered species by identifying genetic risk factors for cheetahs and their offspring.

Project Proposal

1. Describe the overall project and its significance for conservation and/or management. What species, sites, habitats, populations, or questions are targeted or addressed and what critical needs will be met? Explain the project's relevance to the SAFE species' program team's priorities and how it supports implementation of the team's 3-year program plan and existing regional and/or global recovery plans. **LIMIT 5,000**

CHARACTERS with spaces

Since 1981, perplexed zoo veterinarians and animal curators have documented abysmal sperm quality in populations of cheetahs (*Acinonyx jubatus*) (Cohn 1986). The first samples revealed that up to 70% of sperm cells are malformed and sperm concentration is 10% that of domestic cats. This phenotype, teratospermia, is widespread across wild and ex situ cheetah populations (Terrell et al. 2016). The IUCN classifies cheetahs as "Vulnerable" (and "Critically Endangered" in part of their range), so they have been intensively managed in zoos. Even so, cheetah breeding attempts frequently fail to produce cubs: only 20% of planned breeding events per year are successful (Crosier et al. 2021).

Teratospermia is widespread in captive populations of endangered species, including other felines and high-profile conservation species such as black-footed ferrets (Fitzpatrick and Evans 2009; Pukazhenthi et al. 2006; Santymire et al. 2018). Understanding the biological reasons for these animals' reproductive failure is essential to building healthy captive populations of endangered species. Teratospermia is known to have a genetic basis, but that basis is not fully understood.

In 2015, two studies investigated the causes of cheetah teratospermia. Dobrynin et al. sequenced genomes of seven wild-born cheetahs and scanned for single nucleotide polymorphism (SNP) depletion, using cat, dog, and human genomes as controls (Dobrynin et al. 2015). The study identified 18 sperm production and function genes that carry common deleterious mutations in cheetahs. One, AKAP4, was confirmed to have cheetah-specific amino acid substitutions with putative fertility-reducing effects. Dobrynin et al. observed that these impaired genes are strong candidates to explain cheetahs' teratospermia but did not create SNP profiles across a sample of the population, nor did they associate any specific SNPs with aspects of the phenotype.

Koester et al. investigated the influence of human interaction on cheetah sperm production (Koester et al. 2015). They concluded that off-exhibit males produced two-fold more motile sperm than on-exhibit counterparts. These results suggest that the teratospermia phenotype is affected by environmental stressors and is under epigenetic, gene regulatory, or hormonal control. Dobrynin's and Koester's studies inspired my research question: What is the genetic basis for teratospermia in cheetahs?

This project will examine, at the genetic and epigenetic levels, the biology underlying teratospermia in cheetahs. We will conduct whole genome and reduced representation bisulfite sequencing (for DNA methylation) on 30 male cheetahs from the North American ex situ population, including animals held on- and off-exhibit. Single nucleotide polymorphisms with functional effects in fertility-influencing sequences will be identified; differential DNA methylation between study groups will be quantified. The domestic cat will be used as a control.

This study aligns well with a draft goal of the cheetah SAFE program team, which is to maintain genetics and health in the globally managed population. Understanding reproductive health is essential to maintaining a stable ex situ population (Thongphakdee et al. 2020), and management of genetics is an important part of both population stability and reproductive health (Segelbacher et al. 2022; Theissinger et al. 2023). Previously, genome sequencing of endangered species has provided insight into population demographic history and management recommendations in other felid species (e. g., Abascal et al. 2016; Saremi et al. 2019). Accordingly, cheetahs' population-wide genetic similarity has historically been of interest in conservation (Cohn 1986; Terrell et al. 2016; O'Brien et al. 1985). Management of the captive cheetah population at the genetic level has been successful in avoiding loss of genetic diversity (Terrell et al. 2016), demonstrating the effectiveness of the technique.

The examination of epigenetics, another emerging tool in conservation (Rey et al. 2019), will add strength to our conclusions. Epigenetics, as the study of reversible biochemical changes regulating gene expression, provides an often-overlooked biological route for environment to affect phenotype. Changes in epigenetics can occur quickly and are sometimes associated with adaptive phenotypes, which can help sustain small populations lacking genetic diversity. In cheetahs, the apparent contradiction exists of high homozygosity levels and genetic variant fixation and the ability of a genetically based phenotype to vary in response to the environment. An epigenetic influence in this phenotype would provide a plausible explanation.

This study will create a SNP panel of variants associated with male infertility for which all North American SSP cheetahs can be inexpensively screened. This assessment will help lead to a healthier population by identifying factors that inhibit conservation breeding.

2. List the goal(s) and associated objective(s) of the proposed project and the actions that will be taken to achieve them. Goals should describe the desired status in the long-term of the question or problem that the project will address. Objectives should state the immediate results that are necessary to meet the long-term goal(s). Actions should be the specific activities the project will undertake to complete each objective. If the proposal is a request to fund part of a larger project, this section should focus on the part for which CGF funding is being requested during the one-year duration, while also providing some context regarding the larger effort. **LIMIT 4,500 CHARACTERS with spaces**

Objectives:

- 1) Identify deleterious single nucleotide polymorphisms (SNPs) that are associated with phenotypic aspects of teratospermia. Actions include the following:
 - Whole genome sequencing from *Acinonyx jubatus* whole blood samples
 - Characterization of sperm phenotypes (sperm count, % motile, etc.) of study animals – some may already be known from previous studies
 - Identify fertility-related genes
 - Identify SNPs in fertility-related genes
 - Identify SNP frequencies, effects, and conservation scores
 - Identify associations between SNPs and teratospermia phenotypes
 - 2) Identify DNA methylation status of teratospermia-associated loci and associate methylation status with phenotype
 - Reduced representation bisulfite sequencing in semen samples from study cohort
 - Identify methylation status of loci identified in Objective 1
 - Identify functional effects of methylation
 - Compare cheetah methylation frequencies with control (domestic cat) methylation frequencies
 - Compare overall methylation frequencies of on-exhibit and off-exhibit groups; identify associations
 - Compare methylation frequencies between phenotypic groups from Objective 1; identify associations
 - 3) Design a SNP genotyping assay that can be used to screen cheetahs for deleterious variants associated with teratospermia
3. Explain in detail the methodology for this project and why it is appropriate. If this is a research project, clearly state the research question(s) and how it will be tested. For applied conservation projects, explain how project actions will be undertaken and why the proposed actions are appropriate for addressing the conservation issue identified. If this is a conservation education or professional training project, describe the methods as they relate to the target audience, number of people that will be reached, etc. **LIMIT 7,000 CHARACTERS with spaces**

Our research question is: “What is the genetic and epigenetic basis for teratospermia in cheetahs?” This question breaks down into two parts: “What SNPs, if any, are associated with teratospermia?” and “What sites of differential DNA methylation, if any, influence the phenotype of teratospermia?” Each of these questions will be tested using next-generation sequencing and association studies, an essentially hypothesis-free method.

Whole genome sequencing (WGS)

Whole genome sequencing will be outsourced to a local genomics laboratory. This approach is more time-efficient for project staff and results in high-quality data. We will use whole blood samples from male cheetahs in the North American ex situ population banked at the [REDACTED] Institute [REDACTED]

[REDACTED] Our study cohort will include animals held on- and off-exhibit, with a family group in each category. New sample collections will not be required or will be performed during standard veterinary checkups, so permits will not be required.

Illumina paired-end sequencing will be conducted on all samples according to standard protocols (Illumina 2023). Genomic DNA is extracted from whole blood samples using the DNeasy Blood and Tissue Kit (Qiagen) and eluted into 200 µL of Tris-EDTA buffer; extracted genomic DNA will be quantified using the dsDNA High Sensitivity Qubit assay (Invitrogen). DNA is sheared into 350 bp fragments using ultrasonication. Fragments are ligated to standard adapters and index sequences and amplified to create a genomic library using the Illumina TrueSeq DNA PCR-Free library kit. The library will be quantified and quality checked before being sequenced on an Illumina NovaSeq 6000 sequencer to a depth of 20x coverage.

Read data will be processed using the PALEOMIX pipeline. Quality checks will be conducted, datasets cleaned of poor-quality reads, and adapters trimmed from sequences. All computer analyses will be conducted using the Smithsonian Institution's high-performance computing cluster, HYDRA, to which all investigators have access. Processed reads will be mapped to the chromosome-length, high-contiguity cheetah reference genome (Winter et al. 2023). Variants will be called using the GATK best practices workflow (Van der Auwera and O'Connor, 2020) and subsequently filtered using vcftools (Danecek et al. 2011).

We will use two existing online databases, the Male Infertility Knowledgebase (Joseph and Mahale, 2021) and the Infertility Disease Database (Wu et al. 2021), to identify orthologs of fertility-related genes in the cheetah. SNPs will be called from these genes, with the domestic cat (*Felis catus*) as a control. Each SNP's frequency, effects (benign vs. damaging) and site conservation score will be calculated.

Many software applications exist to predict SNP functional effects (Cline & Karchin, 2011; Bertorelle et al. 2022). They can assess whether an SNP is in a coding region or a noncoding region and check its function accordingly, answering questions such as: would it change an amino acid in a protein? What functional effect would that change have? Is the SNP in a regulatory region? Does it affect transcription factor or miRNA binding? We will use the tools SnpEff v5.1 and SnpSift to assess the probable effects of our SNPs (Cingolani et al. 2012a, b). This analysis will be essential to determining whether a SNP plays a role in the teratospermia phenotype.

We will also calculate site conservation scores across species. High conservation scores indicate that a sequence is similar or identical across species. Usually, high scores indicate that the site is functionally important (Siepel et al. 2005). The program phastCons uses a hidden Markov model to analyze site conservation scores across the phylogenetic tree and has produced high-quality tables of sequence conservation in eukaryotes (Siepel et al. 2005). This program is now available through a web interface, PhastWeb, which we can use to calculate site conservation scores among felid species and in less related eukaryotes (Ramani et al. 2019).

We will characterize the semen phenotypes of our study males. Sperm concentration, total sperm, percent motile, total motile, and percent normal will be measured. We will test the statistical association of SNP presence with aberrant phenotype (numbers outside of normal range) in each category using t-tests and ANOVA. Finally, putatively damaging SNPs associated with teratospermia will be analyzed with snps2assays (Anderson, 2015) to design a SNPtype assay for use with the Fluidigm Biomark system.

Reduced representation bisulfite sequencing (RRBS)

We will conduct reduced representation bisulfite sequencing on matched semen samples from our study cohort. This method is a gold-standard technology for epigenetic sequencing because it identifies DNA methylation at single-base resolution and its data can be used both quantitatively and qualitatively (Li et al. 2011). DNA from sperm cells in semen will be lysed using the modified protocol, "Purification of total DNA from animal sperm using the DNeasy Blood & Tissue Kit; protocol 2" (Qiagen). Library preparation and sequencing will be outsourced. Methylation libraries will be prepared using the protocol of Meissner et al. (2005), modified for use with the Zymo-Seq RRBS Library Kit. In RRBS, genomic DNA is digested using restriction enzymes. DNA fragments will be ligated to adapters; excess adapter will be removed and preparative electrophoresis performed. Adapter-ligated fragments will be bisulfite-converted, which deaminates all unmethylated cytosine nucleotides into uracil. Methylated cytosines do not deaminate. Adapter-ligated fragments will be enriched via PCR and products subsequently cleaned and eluted. Libraries from multiple individuals will be pooled and sequenced on an Illumina NovaSeq 6000 sequencer.

Sequence read processing follows essentially the same protocol as for WGS data, above. BS-Seeker3 (Huang et al. 2018) and bowtie2 (Langmead & Salzberg 2012) will be used to construct an RRBS index from the cheetah reference genome. Because unmethylated cytosines were converted into uracil, remaining cytosines indicate sites of DNA methylation.

We will determine the methylation frequency, measured as the frequency of Cs, and methylated sites within each fertility-influencing gene. Methylation variable positions (MVPs) and differentially methylated regions (DMRs) will be identified (Lea et al. 2015). Binomial mixed models will be used to compare methylation frequencies at these sites between males held on-exhibit, males held off-exhibit, and the domestic cat control. Frequencies will also be compared among groups with different teratospermia phenotypes. We will also assess the degree of epigenetic inheritance among the family group included in each exhibit status cohort.

4. If AZA has provided previous funding for this or a related project through its grants programs, summarize relevant findings and identify how they relate to the current proposal. **LIMIT 1,000 CHARACTERS with spaces**

AZA has never provided funding for this project. However, it has funded other projects based on genotyping to improve reproductive success. Calatayud et al. at the San Diego Zoo Institute for Conservation Research were awarded a grant for "Applying SNP-Derived Genotyping to the Captive Management of the Critically Endangered Mountain Yellow-Legged Frog." They used next-generation sequencing techniques to determine the relatedness of founder individuals for their captive population and discovered that the founders were more closely related than they supposed, which changed

that population's breeding recommendations. Our proposed research, like theirs, seeks to refine breeding recommendations using genetic data and NGS. AZA has also funded previous research into the effects of male physiology on reproductive success and sustainability in felids, such as its CGF grant to Herrick et al. on snow leopards (*Panthera uncia*) in 2015.

5. Describe the timeline in which components of the project will be completed within the one-year project duration. Use of bulleted/numbered lists is encouraged. **LIMIT 1,500 CHARACTERS with spaces**

Nov 2023-Dec 2023:

- o Submit whole blood samples for WGS. 4-6 weeks.
- o Identify fertility-related genes in cheetah reference sequences.

Jan 2024-Feb 2024:

- o WGS read processing and assembly.
- o Semen phenotype characterization of study animals.

Mar 2024-Apr 2024:

- o Analyze whole genome sequences.
- o Identify SNP sites, frequencies, effects, and conservation scores.
- o Submit semen samples for RRBS. 4-6 weeks.

May 2024-Jun 2024:

- o Process, assemble, and index RRBS data.

Jun 2024-Aug 2024:

- o Analyze RRBS data.
- o Identify methylation sites, DMRs, and MVPs.
- o Calculate differences among exhibit status and phenotypic groups.

Sep 2024-Oct 2024:

- o Design SNP genotyping assay to screen cheetahs for deleterious variants associated with teratospermia.
- o Present at international conferences, zoos, and colleges.
- o Submit final report to AZA.

6. Define the project team by identifying the personnel included for this project, their roles, qualifications, and the key skills they bring. Discuss the PI and each CI listed on the Cover Sheet and any collaborators (persons actively involved in the completion of the project). Each collaborator's institution should be listed under **Collaborating Institutions or Organizations** on the application Cover Sheet. **LIMIT 2,500 CHARACTERS with spaces**

[REDACTED]

[REDACTED] He will help supervise and participate in the data analyses, manuscript preparation, and preparation of reports to AZA.

[REDACTED]

[REDACTED] Her experience leading and advising effective management programs makes her an ideal person to identify and implement any conservation and management actions that our results suggest.

[REDACTED]

[REDACTED] She has experience in zoo education and collegiate teaching, which prepare her well for the educational outreach activities she is planning.

For advice on the epigenetics aspect of this project, we will collaborate with [REDACTED] an epigenetics expert at [REDACTED]. Her current research focuses on semen epigenetics in black-footed ferrets.

7. Describe the project's monitoring and evaluation plans. Explain how the goals and objectives of the project will be met, which indicators will be used, and how those indicators will be measured. If this project has an educational or professional training component, the educational impact should be discussed (ideally including impacts on conservation-related knowledge, attitudes/affect, and behavior). Plans may be quantitative or qualitative, and should differentiate between monitoring (ensuring proposed actions are taken) and evaluation (whether the project had the desired impact on the issue or question addressed).

LIMIT 3,000 CHARACTERS with spaces

For Objective 1, the indicator will be the presence of functionally effective SNPs whose presence statistically correlates with one or more aspects of the teratospermia phenotype. Monitoring activities will be data analysis and statistical tests on WGS data. [REDACTED] will perform these analyses by April 2024.

Measures of success will include:

- Isolate high-quality genomic DNA from whole blood samples from 30 cheetahs and 1 domestic cat.
- Complete whole genome sequencing to 20x coverage on all samples.
- Align sequence reads and call and analyze variants/SNPs using the high contiguity reference genome assembly of the cheetah.
- Localize variants linked to teratospermia and evaluate their potential impact.
- Share results with AZA Felid TAG
- Prepare manuscript of results and submit to peer-reviewed scientific journal.

For Objective 2, indicators will be the number of SNPs from Objective 1 with known methylation status and whose methylation status statistically correlates with one or more aspects of the teratospermia phenotype. Monitoring activities will be analysis and statistical tests on methylation data. [REDACTED] will perform these tests by October 2024.

Measures of success will include:

- Isolate high-quality genomic DNA from semen samples from 20 cheetahs and 1 domestic cat.
- Complete reduced representation bisulfite sequencing, generating at least 25 million reads per sample.
- Align sequence reads to reference genome.
- Analyze methylation frequencies, DMRs, and MVPs at male fertility-related genes.
- Share results with AZA Felid TAG.
- Prepare manuscript of results and submit to peer-reviewed scientific journal.

For Objective 3, we will design a cheetah-specific genotype assay of SNPs identified in Objective 1 that can be used to screen the AZA cheetah SAFE population. [REDACTED] and [REDACTED] will design this assay by November 2024.

8. Explain and identify the means and timeline of how the key results and lessons of the project will be disseminated and shared. Key internal and external audiences should be identified. **LIMIT 1,500 CHARACTERS with spaces**

The WGS data we generate will be uploaded to the Sequence Read Archive at NCBI's GenBank database. We will publish our findings in a peer-reviewed research journal. CI [REDACTED] will incorporate this research into her Ph.D. dissertation, which will be publicly defended and submitted to the [REDACTED] University library. The work will be available to faculty and the 38,000-person student body. Results will be presented to the AZA Felid TAG. [REDACTED] also plans to present educational talks on this project. Potential locations include scientific conferences, such as the AZA Annual Meeting, the North American Congress for Conservation Biology, and the International Plant & Animal Genome Conference. She has also been invited to speak at [REDACTED], a small liberal arts college, and the [REDACTED], an intensive conservation training environment. She will present her findings as part of a zoo education program at the [REDACTED] Zoo.

9. Have all the necessary permits been obtained for the project?

a) ☐ Yes ☐ No ☒ Not Applicable

If yes, list the permits here and attach one copy of each to your Application email. Attachments should be titled "Permit #."

b) If the necessary permits have not yet been obtained, please explain why. Pending permits must be obtained and submitted to AZA by 15 June 2023.

10. If research animals are to be purchased for the project, indicate and justify the destination of the animal(s) after completion of the research. **LIMIT 750 CHARACTERS with spaces**

11. If this is a multi-year project, identify the funding strategy for subsequent years. **LIMIT 750 CHARACTERS with spaces**

12. **Past AZA Grant Support.** If Principal or Co-Investigators listed on the Cover Sheet have received previous AZA grant support (i.e., CGF or SAFE granting program), complete the following (required).

a) Name of Investigator: [REDACTED] Year of Award: 2021 Amount of Award: \$18,758

Project Title: [REDACTED]

Were all required reports (annual/final) submitted?

☐ Yes ☐ No ☒ Not yet due / Other (briefly explain) This award is expiring on 03/31/2023.

b) Name of Investigator: Year of Award: Amount of Award:

Project Title:

Were all required reports (annual/final) submitted?

☐ Yes ☐ No ☐ Not yet due / Other (briefly explain)

Note: If additional awards need to be included, add their information to a separate document titled "Past AZA-Funded Projects" and include it as an attachment with your Application email.

2023 AZA CONSERVATION GRANTS FUND

Project Budget

BUDGET CYCLE

All funding requests from CGF must be for a maximum period of one year only. Funding decisions are announced in early September and expenses incurred prior to the award date will not be covered by CGF funds. CGF funds are to be used within one year from the project start date and forward.

DETAILED BUDGET

A detailed budget sheet is provided below and information should be separated into CGF and non-CGF costs. List under each category the budget items and method of calculation for each (e.g., 2 persons x 10 days x \$25/day), as well as probable sources of support. The following should be listed as **separate** budget items:

- airfares
- lodging
- vehicle rental (automobiles, planes, boats)
- fuel
- per diem
- equipment over \$500
- supplies, services (lab analyses, editing)
- pharmaceuticals
- shipping/postage
- printing
- personnel costs

CGF does not fund the following: Tuition, university fees, or fringe benefits associated with graduate students or other project personnel; administrative costs (including institutional overhead and submission costs for the publication of journal articles); travel unrelated to completion of the project (conference presentation expenses, travel to meetings, etc.); planning meetings that define, rather than implement, conservation goals.

CGF does not fund salaries for project personnel other than graduate students, technicians, and start-up positions established solely for the purpose of completion of the project.

Note: Recipients of CGF grants are obligated to inform AZA (CGF@aza.org) of any pending or overlapping grants received. If such grants are received, AZA's Scientific Advisory Committee may recommend that the award amount be reduced or refunded.

Note: If additional budget items need to be included, add this information in a separate document titled "**Additional Budget Items**" and include it as an attachment with your Application email.

Budget Items Form:

[illegible]

	TOTAL PROJECT BUDGET (round to nearest whole dollar)	\$23,300	\$			\$23,300
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BUDGET JUSTIFICATION

State why each item listed in the Budget Items Form costs what it does and indicate how each item relates to the proposed scope of work. Sufficient detail should be provided to address any potential reviewer concerns with respect to cost and need. If the lifespan of equipment costing over \$500 exceeds that of the grant period, detail and justify who will assume ownership of the item. **LIMIT 2,500 CHARACTERS with spaces**

We request \$23,300 to conduct whole-genome and reduced representation bisulfite sequencing on 50 samples from 30 cheetahs at 20x coverage. Sequencing will be carried out by commercial vendors, according to the attached price quotes. We will conduct sample extraction ourselves, saving \$1950 based on the given quote.

Funding is requested to purchase extraction supplies, including 2 Qiagen DNeasy Blood and Tissue Kits.

Funding is requested for whole-genome sequencing of DNA isolated from whole blood samples at 20x coverage. Sequencing 30 samples will enable us to create a meaningful SNP profile and add statistical power to our analyses. The 31st sample is the control.

Funding is requested for reduced representation bisulfite sequencing of DNA isolated from matched semen samples at 20x coverage. We will need to sequence 20 samples for this project. The 21st sample is the control.

Funding is also requested for whole genome sequence data delivery via an external hard drive.

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Below is the price quote for WGS and RRBS from Psomagen, Inc. (the lowest of the 3 quotes we received):

Study 1 - ANQ00002573

- o Application: whole genome sequencing
- o Species: animal (cheetah / *Acinonyx jubatus*)
- o Starting material: per sample ~2mL of whole blood
- o Sample size: 30
- o Library kit: TruSeq DNA PCR-Free Library kit
- o Sequencing platform: NovaSeq6000 S4 (150PE)
- o Output: per sample ~50Gb raw throughput
- o Deliverables: raw data (fastq) only
- o Data delivery method: external harddrive (2TB), 1 unit
- o Estimated price:

	Unit Price	Unit	Sub Total
Extraction	\$35.00	30	\$1,050.00
Library Prep & Sequencing	\$550.00	30	\$16,500.00
Data Handling and Shipping	\$150.00	1	\$150.00
Total Price			\$17,700.00

- o Turnaround time: 4-6 weeks after extracted samples pass their initial QC

Study 2 - ANQ00002574

- o Application: Reduced Representation Bisulfite Sequencing (RRBS)
- o Species: animal (cheetah / *Acinonyx jubatus*)
- o Starting material: sperm cells / gDNA isolation is needed
- o Sample size: 20
- o Library kit: Zymo-Seq RRBS Library Kit
- o Sequencing platform: NovaSeq6000 S4 (150PE)
- o Output: per sample ~40M paired reads (20M read1 and read2)
- o Deliverables: raw data (fastq) only
- o Data delivery method: sFTP download
- o Estimated price:

	Unit Price	Unit	Sub Total
Extraction	\$45.00	20	\$900.00
Library Prep & Sequencing	\$270.00	20	\$5,400.00
Total Price			\$6,300.00

- o Turnaround time: 4-6 weeks after extracted samples pass their initial QC

